IAEA

The International Atomic Energy Agency (IAEA) was established in 1957 as an autonomous international organization within the United Nations system, following growing international recognition of both the benefits and dangers associated with nuclear technology. Created under the principle of “Atoms for Peace,” the Agency was given the responsibility of promoting the peaceful use of nuclear energy while helping to prevent its diversion toward military purposes.Over the decades, the IAEA's responsibilities expanded significantly to include nuclear safety, nuclear security, safeguards, technical cooperation, radiation protection, and the management of radioactive waste. As nuclear technology became increasingly important in electricity generation, medicine, agriculture, industry, and scientific research, the safe handling of radioactive materials became a major international concern.

The International Atomic Energy Agency (IAEA) is an international organization focused on promoting the peaceful, safe, secure, and responsible use of nuclear technology. The Agency assists Member States in developing nuclear energy programmes, establishing effective safety frameworks, managing radioactive materials, and ensuring that nuclear technologies contribute to sustainable development. As global interest in nuclear energy continues to increase, the IAEA plays an increasingly important role in addressing the challenges associated with spent nuclear fuel, radioactive waste, and the long-term environmental responsibilities of nuclear power generation.

Topic 1 -

The Expansion of Nuclear Energy and Ensuring Safe Management of Radioactive Waste

Topic Agenda

The increasing global demand for reliable, low-carbon energy has renewed international interest in the expansion of nuclear power. Nuclear energy has the potential to provide large quantities of electricity with relatively low operational carbon emissions, leading several states to consider constructing new nuclear power plants, extending the operational life of existing reactors, and developing emerging technologies such as Small Modular Reactors (SMRs).

However, the expansion of nuclear energy also raises significant concerns regarding the management of radioactive waste and spent nuclear fuel. Radioactive waste is produced not only through nuclear power generation but also through the use of radioactive materials in medicine, industry, agriculture, and scientific research. Depending on its characteristics, certain radioactive waste may remain hazardous for extremely long periods and therefore requires careful treatment, storage, transportation, and disposal.While significant scientific progress has been made in radioactive waste management, challenges remain regarding the establishment of long-term disposal facilities, public acceptance, financing, international cooperation, and the differing technological capabilities of Member States. Some countries have developed advanced strategies for geological disposal and long-term waste management, while others continue to rely primarily on interim storage facilities.The countries must therefore consider how

nuclear energy can expand responsibly without transferring disproportionate environmental and safety burdens to future generations.

This topic encourages delegates to examine the balance between energy security, sustainable development, nuclear safety, and environmental protection while exploring international frameworks that promote the safe and responsible management of radioactive waste.

Present Issues

Global interest in nuclear energy is increasing as states seek reliable energy sources capable of supporting economic development while reducing greenhouse gas emissions. However, every stage of the nuclear fuel cycle produces responsibilities that extend beyond the operation of a reactor. The management of spent fuel and radioactive waste remains one of the most complex long-term challenges associated with nuclear technology.Radioactive waste varies significantly in its level of radioactivity and longevity. Low-level waste can often be managed and disposed of through established methods, while intermediate- and high-level waste may require more sophisticated containment and long-term isolation. Spent nuclear fuel presents particular challenges due to its high radioactivity and the extended periods for which it must be safely managed.

The IAEA has developed international safety standards and supports Member States in developing national radioactive waste management programmes. The Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management remains the principal legally binding international instrument specifically addressing safety in these areas. Nevertheless, significant differences exist between states regarding technological capacity, funding, regulatory infrastructure, geological conditions, and public acceptance of disposal facilities.One of the most significant obstacles to establishing long-term radioactive waste disposal programmes is public confidence. Communities may oppose the construction of storage or geological disposal facilities because of concerns regarding environmental contamination, transportation accidents, and the long-term consequences of radioactive materials. The IAEA has identified public perception, inadequate funding mechanisms, and historical waste-management practices among the challenges that can delay or obstruct disposal programmes.

The rapid development of new nuclear technologies also creates additional regulatory challenges. Small Modular Reactors and other emerging reactor technologies could allow nuclear energy to become accessible to a wider range of countries. However, the expansion of nuclear programmes requires corresponding investments in regulatory capacity, waste management infrastructure, transportation safety, and long-term disposal strategies.As more states consider nuclear power as part of their future energy strategies, the international community faces an important question: How can the benefits of nuclear energy be expanded while ensuring that radioactive waste is managed safely, responsibly, and sustainably for present and future generations?.

Key Issues Delegates Have to Address

Determining appropriate methods for the long-term disposal of high-level radioactive waste and spent nuclear fuel, including the development of deep geological repositories and other internationally accepted disposal solutions.

Safe Management of Spent Nuclear Fuel
Establishing effective systems for the storage, transportation, treatment, recycling, and eventual disposal of spent nuclear fuel while minimizing risks to human health and the environment.

International Safety Standards
Strengthening international cooperation and encouraging Member States to implement IAEA safety standards and effective national regulatory frameworks for radioactive waste management.

Financing Radioactive Waste Management
Determining how the costs associated with the storage, transportation, disposal, and long-term monitoring of radioactive waste should be funded without placing disproportionate financial burdens on future generations.

Nuclear Energy Expansion and Infrastructure
Ensuring that states expanding their nuclear energy programmes possess adequate infrastructure, technical expertise, regulatory institutions, and waste management strategies before constructing new nuclear facilities.

Protection of Human Health and the Environment
Developing measures to prevent radioactive contamination of ecosystems, groundwater, communities, and future generations throughout the entire lifecycle of radioactive materials.

Transportation and Cross-Border Cooperation
Addressing the safe transportation of radioactive materials between states and developing mechanisms for international cooperation while respecting national sovereignty and environmental concerns.

Public Confidence and Community Participation
Improving transparency, public communication, and community involvement in decisions concerning radioactive waste storage and disposal facilities.

Technological Innovation
Encouraging research into advanced waste treatment, recycling, storage, and disposal technologies that could reduce the volume, toxicity, or long-term hazards associated with radioactive waste.

Support for Developing Nuclear Programmes
Providing technical assistance and international cooperation to states with limited resources that are developing nuclear energy programmes and radioactive waste management infrastructure.

Long-Term Responsibility
Establishing principles regarding the responsibility of current generations and governments to safely manage radioactive waste without transferring unacceptable risks or obligations to future generations.

Security of Radioactive Materials
Preventing the loss, theft, unauthorized access, or misuse of radioactive materials during storage, transportation, and disposal.Long-Term Disposal of Radioactive Waste

Guiding Questions

How can states ensure that radioactive waste management plans are established before the expansion of nuclear energy programmes?

What international standards should govern the long-term storage and disposal of high-level radioactive waste and spent nuclear fuel?

How can the international community support developing states that lack the financial or technological capacity to manage radioactive waste independently?

Should greater international cooperation be established for shared radioactive waste management facilities, and how can issues of sovereignty and environmental responsibility be addressed?

What measures can be implemented to improve public confidence and transparency regarding radioactive waste disposal facilities?

How can new technologies such as Small Modular Reactors be developed while ensuring that appropriate waste management strategies are established?

Who should bear the long-term financial responsibility for radioactive waste management and nuclear facility decommissioning?

How can the IAEA strengthen international cooperation and encourage Member States to adopt effective radioactive waste management policies?

Recommended Sources

IAEA – Radioactive Waste and Spent Fuel Management

IAEA – The Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management

IAEA – Managing Radioactive Waste

IAEA – Status and Trends in Spent Fuel and Radioactive Waste Management

IAEA Safety Standards – Disposal of Radioactive Waste (SSR-5)

IAEA – Communication and Stakeholder Involvement in Radioactive Waste Disposal

IAEA – Roadmap for Implementing a Geological Disposal Programme

United Nations Treaty Collection – Joint Convention on Radioactive Waste Management

Topic 2 -

Nuclear Security in the Age of Artificial Intelligence and Cyber Warfare

Topic Agenda

As nuclear facilities become increasingly dependent on digital technologies, the nature of threats facing the global nuclear sector is rapidly evolving. Nuclear power plants, research reactors, fuel-cycle facilities, and other institutions handling radioactive materials rely on complex computer systems to support monitoring, communications, security, and operational processes. Although digitalization offers significant improvements in efficiency and safety, increased technological dependence also exposes critical infrastructure to emerging cyber threats.Cyberattacks against nuclear facilities have the potential to compromise sensitive information, disrupt essential services, interfere with operational systems, or undermine public confidence in nuclear safety. The emergence of increasingly sophisticated artificial intelligence technologies has further transformed the cybersecurity landscape. AI may enable malicious actors to identify vulnerabilities, automate cyber operations, and develop more complex methods of digital intrusion. Conversely, these same technologies can strengthen nuclear security through improved threat detection, system monitoring, predictive analysis, and rapid incident response.The challenge facing the international community is therefore not only preventing cyberattacks but also ensuring that all states possess the capacity to protect increasingly interconnected nuclear infrastructure. Significant differences exist between Member States regarding technological capabilities, cybersecurity expertise, regulatory frameworks, and access to advanced security systems. Additionally, the sensitive nature of nuclear information creates difficulties in establishing effective international cooperation and information-sharing mechanisms.

This topic invites delegates to examine the intersection between artificial intelligence, cybersecurity, and nuclear security. Delegates are encouraged to explore international strategies that strengthen the protection of nuclear facilities, establish common cybersecurity standards, improve cooperation between Member States, and promote the responsible application of artificial intelligence in nuclear security. In

doing so, delegates must balance the need for stronger international coordination with national sovereignty, confidentiality, technological development, and the continued peaceful use of nuclear technology.

Present Issues

The growing dependence of nuclear facilities on digital systems has significantly increased concerns surrounding cyberattacks and emerging AI-enabled threats. Modern nuclear infrastructure relies on interconnected instrumentation, control, and information systems, creating potential vulnerabilities that could affect nuclear safety and security if compromised. The IAEA has emphasized the importance of protecting these systems against cyberattacks, manipulation, and falsification, while discussions at the 2024 International Conference on Nuclear Security highlighted the growing need for continuous monitoring and cybersecurity assessments as autonomous systems become more widely integrated. In 2024 alone, states reported 147 incidents involving nuclear and other radioactive material outside regulatory control, demonstrating the continuing importance of strengthening global nuclear security mechanisms.

Artificial intelligence presents both opportunities and risks for the nuclear sector. While AI can improve threat detection, anomaly monitoring, and cybersecurity response, malicious actors may also use increasingly advanced technologies to automate cyber operations and identify vulnerabilities more rapidly. A major challenge remains the disparity between Member States: technologically advanced countries may possess specialised cybersecurity capabilities, while states developing nuclear programmes may lack sufficient infrastructure and trained personnel. The IAEA continues to expand capacity-building efforts, including specialised computer security training, but the rapid evolution of cyber and AI technologies creates an urgent need for stronger international cooperation and consistent standards to protect critical nuclear infrastructure.

Key Issues Delegates has to address

Cybersecurity of Nuclear Facilities
Developing stronger cybersecurity measures to protect nuclear power plants, research reactors, and radioactive material facilities from cyberattacks and unauthorized access.

AI-Enabled Cyber Threats
Addressing the potential misuse of artificial intelligence to identify vulnerabilities, automate cyberattacks, or disrupt critical nuclear infrastructure.

Protection of Critical Digital Systems
Ensuring that computer-based monitoring, control, communication, and safety systems remain secure and operational during cyber incidents.

International Cybersecurity Standards
Determining whether the IAEA should establish stronger international standards or guidelines for cybersecurity within nuclear facilities.

Information Sharing Between States
Creating mechanisms for Member States to share information about cyber threats and incidents while protecting sensitive national security information.

AI for Nuclear Security
Establishing guidelines for the responsible use of artificial intelligence in threat detection, cybersecurity monitoring, emergency response, and nuclear facility protection.

Protection Against State-Sponsored Cyberattacks
Addressing the possibility of governments or state-sponsored groups targeting nuclear infrastructure as part of geopolitical conflicts.

Cybersecurity Capacity of Developing States
Providing technical assistance, training, and resources to states that lack the expertise or infrastructure necessary to adequately protect their nuclear facilities.

Incident Response and Recovery
Developing international mechanisms to help nuclear facilities detect, contain, and recover from cyberattacks while maintaining nuclear safety.

Balancing Security and Sovereignty
Determining how international cybersecurity cooperation can be strengthened without requiring states to disclose sensitive information or surrender control over their national nuclear infrastructure.

Preventing Unauthorized Access
Strengthening authentication, access-control systems, and personnel security to prevent unauthorized individuals from gaining access to critical nuclear systems.

International Cooperation on Emerging Technologies
Encouraging states to cooperate on cybersecurity research and AI development while preventing advanced technologies from being exploited against nuclear infrastructure.

Guiding Questions

To what extent should the IAEA establish international cybersecurity standards specifically for nuclear facilities?

How can Member States protect nuclear infrastructure from increasingly sophisticated cyberattacks and AI-enabled threats?

What measures should be implemented to prevent artificial intelligence from being misused against nuclear facilities?

How can AI be responsibly utilized to improve nuclear cybersecurity, threat detection, and emergency response?

How can states share information about cyber threats and incidents while protecting sensitive nuclear security information?

What assistance should the IAEA provide to developing countries that lack the technical expertise and resources to protect their nuclear infrastructure?

Should states be required to report significant cyber incidents affecting nuclear facilities to the IAEA?

How can nuclear facilities maintain safe operations and recover quickly following a successful cyberattack?

What international measures can prevent state-sponsored cyberattacks against nuclear infrastructure from escalating into wider conflicts?

How can the international community balance national sovereignty, cybersecurity, transparency, and the peaceful use of nuclear technology?

Recommended Sources

IAEA – Computer Security for Nuclear Security (NSS No. 42-G)

IAEA – Computer Security Techniques for Nuclear Facilities (NSS No. 17-T)

IAEA – Nuclear Security Series

IAEA – Nuclear Safety and Security Online User Interface

IAEA – Nuclear Security Information Portal

IAEA – Nuclear Security eLearning: Computer Security

Country List

🇺🇸 USA

🇨🇳 China

🇷🇺 Russia

🇫🇷 France

🇬🇧 UK

🇯🇵 Japan

🇰🇷 South Korea

🇮🇳 India

🇨🇦 Canada

🇩🇪 Germany

🇦🇪 UAE

🇫🇮 Finland

🇸🇪 Sweden

🇹🇷 Türkiye

🇨🇭 Switzerland

🇪🇬 Egypt

🇧🇷 Brazil

🇿🇦 South Africa

🇨🇿 Czech Republic

🇮🇷 Iran

🇦🇷 Argentina

🇰🇿 Kazakhstan

🇸🇰 Slovakia

🇦🇲 Armenia

🇸🇮 Slovenia